When Does Auxiliary Data Help? Global Geometry and Optimal Mixing under Task Mismatch

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Abstract

How should auxiliary data be mixed when target risk has more than one valley? We study this question in a dual-domain full-ridge model with task mismatch. One coordinate cannot produce multiple valleys, while two coordinates suffice: a certified example has two local minima, and a one-parameter family exhibits valley creation, annihilation, and two distinct global minimizers. To identify all global candidates, we reduce the population stationary equation to polynomial root isolation and compare the stationary and boundary risks. Zero mismatch provides a contrasting log-convex or flat regime. We then connect this population geometry to finite samples through a fixed-dimension expansion of the actual risk and its first two derivatives. For sufficiently large samples, a nondegenerate interior optimum admits a first-order weight correction, while two nondegenerate interior population minima are selected by their unequal R_1 values. Under matched spectra and zero mismatch, an exact quadratic R_1 determines the limiting weight when its coefficients are nonzero. Validation combines exact population certificates with conditional Monte Carlo for risk and weight expansions; a fixed-candidate comparison is distinguished from finite-sample branch validation. A general finite-sample shape classification and robot-policy validation remain open.

1. Introduction

Robot demonstrations are expensive, while egocentric human recordings and other auxiliary embodiments are cheap and abundant. Mixed training can exploit this asymmetry, but auxiliary data can also produce negative transfer. Choosing a source fraction requires understanding the target risk across mixing weights: a locally favorable weight need not be globally best, and finite samples can change which candidate is preferred.

We study a dual-domain model with Gaussian covariates, a shared spectral basis, task-vector mismatch δ, and domain-specific noise. Standard full-ridge regression is the main estimator; diagonalized ridge is retained only as an interpretable special case. The model is deliberately simple so that the risk geometry and the global optimum can be characterized rather than inferred from a local derivative or a coarse grid. Closed-loop robot policies are outside the scope of this paper.

Two coordinates already produce multiple valleys. Our starting point is the population risk as a function of the mixing weight. One coordinate cannot produce multiple valleys, whereas two suffice (Proposition 1). A certified two-dimensional example and a one-parameter family exhibit local minima separated by a local maximum, valley creation and annihilation, and two distinct global minimizers. This makes the location and value of every candidate relevant to selecting the mixture. Zero mismatch supplies a contrasting case: log-convexity gives a complete uniqueness and flatness classification.

Algebraic global certification. The population derivative can be cleared of positive denominators to obtain a polynomial stationary equation of degree at most $3p - 2$. For fixed parameters, complete real-root isolation followed by comparison of all stationary and boundary candidates certifies the global optimum without assuming convexity or unimodality. This supplies a direct way to check when the nondegenerate perturbation theorem applies and when a different global-shape analysis is required.

Finite-sample optimal selection. A fixed-dimension expansion of the actual full-ridge risk and its first two derivatives connects these population cases to finite-sample optima (Theorem 3). At a unique nondegenerate interior population optimum, it gives a first-order weight correction and optimized risk and gain (Theorem 4). With exactly two nondegenerate interior population minimizers, the smaller R_1 branch wins for sufficiently large samples. When matched spectra and zero mismatch make R_0 flat,

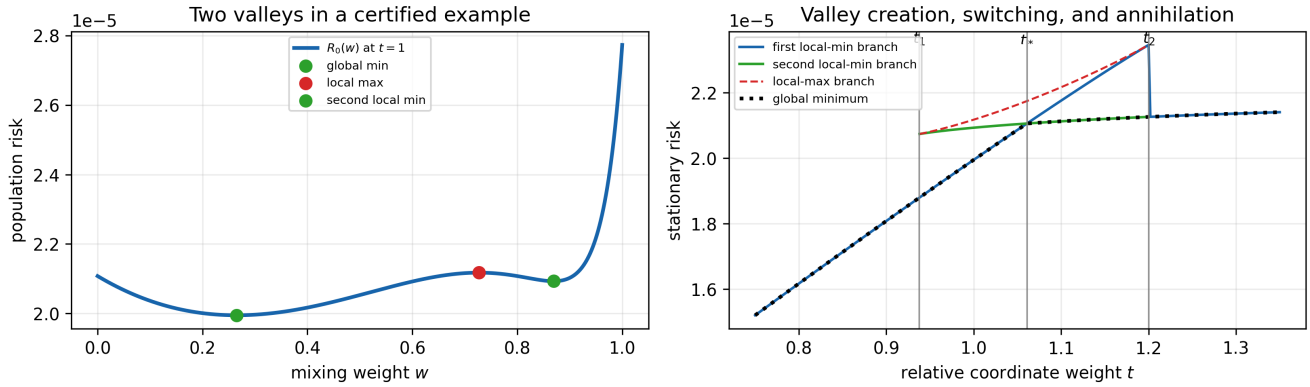


Figure 1. **Global shapes of auxiliary-data mixing risk.** Left: a certified two-dimensional population risk with two local minima and one local maximum; the first minimum is the unique global optimum. Right: stationary branches of a one-parameter family. Valley pairs appear and disappear at t_1, t_2, and at t_* two distinct weights share the global minimum. These are exact algebraic and interval-certified structures, not grid-based shape conjectures.

$R_1 = A(1-w)^2 + Bw^2$ determines the limiting weight and first-order risk and gain if $A + B > 0$. If $A = B = 0$, the risk is exactly zero for every weight. A first-order tie between distinct nontrivial branches is unresolved at this order.

Evidence and scope. Exact root isolation and interval comparisons certify population geometry. Conditional Monte Carlo checks the actual-risk expansion, interior weight corrections, and flat-case gains. The reported comparison at two fixed population weights tests their finite-sample risk ordering; it does not validate persistence or selection of finite-sample local-minimum branches. A general finite-sample shape classification, parameter-estimation guarantees, and robot-policy validation remain open.

Contributions. We analyze global risk geometry and its consequences for choosing a mixing weight: (i) a minimum-dimension multimodality result and a certified two-coordinate family with valley bifurcations and two global minima; (ii) polynomial global certification for fixed parameters, together with a zero-mismatch uniqueness and flatness classification; (iii) a uniform fixed-dimension risk and derivative expansion yielding nondegenerate weight corrections, first-order selection between two population minima, and flat-population optimal selection. Proofs, exact parameters, and numerical protocols are in the supplementary material.

2. Related work

Egocentric human data for robot learning. Egocentric video has long served as a representation prior: Ego4D [7] supplies large-scale first-person video, and R3M/VIP-style pretraining [13, 14] distills it into visual features for downstream control. A more recent line injects egocentric data *directly into policy training*: EgoMimic [10] uses hand-mounted grippers whose kinematics match the robot, UMI [4] uses hand-held grippers with visual-inertial pose recovery, DexCap [17] uses portable mocap, and EgoBridge [16] studies mixing in simulation with explicit domain gaps. Cross-embodiment robot corpora [15] face a cousin of the same problem across robot morphologies. These works demonstrate practical benefits, substitution ratios [5], and failure modes, and motivate our complementary regression-level analysis of how task mismatch and sample scale affect an explicitly optimized mixing weight.

Domain adaptation and transfer regression. Classical domain-adaptation theory bounds target error by source error plus a discrepancy and a shared-optimal-hypothesis term [2]; subsequent work studies when quantity compensates for quality, the value of target data, and when representation invariance can hurt [3, 8, 20]. More closely, [19] analyze two-task linear regression under covariate and model shifts in the proportional high-dimensional limit. They characterize the bias and variance of hard parameter sharing, identify positive-to-negative transfer transitions, and study source rebalancing. [11] consider weighted joint ridge regression with multiple heterogeneous sources, deriving deterministic risk equivalents that capture covariance shift, model shift, sample sizes, regularization, and privacy noise. Their formulation contains pooled weighted ridge as a limiting case and includes Bayes-optimal tuning under Gaussian priors. These works already provide precise transfer-risk analyses and source-weighting principles under task mismatch. We focus on the global geometry along a scalar mixing weight: a minimum-dimension multivalley

construction, certification of all population candidates, and finite-sample selection based on a uniform fixed-dimension risk expansion.

Mixture scaling laws and the present scope. [18] study mixed ridge regression with a shared regression function, heterogeneous covariances and noise levels, and derive minimax and ridge scaling laws under aligned power-law spectra. Sim-to-real co-training analyses provide complementary empirical models of multi-source learning [12]. Relative to this line, we allow an explicit deterministic task mismatch δ and study the geometry and selection of optimal mixing weights at fixed dimension. Scaling exponents in a proportional regime are outside the present analysis.

Technical tools. Our analysis uses ridge-regression risk decompositions [6, 9], with target risk measured in the Σ_r geometry. Real Wishart moments and resolvent expansions retain the joint sample-Gram fluctuations of full ridge. The per-coordinate label-feedthrough calculation belongs to the separate diagonalized estimator in the supplementary. Invariant-representation limitations [1, 20] provide related context; alignment and gradient-conflict experiments are not part of the current main-text validation.

3. A dual-domain model of cross-embodiment mixing

We abstract the problem to its statistical skeleton. Two domains provide training data: the *robot domain* r (target; teleoperated demonstrations) and the *human domain* h (source; egocentric recordings). We write $e \in \{r, h\}$ for the domain index.

Assumption 1 (Data model)**.** Coordinates $j = 1, \dots, p$ index a shared spectral basis. Design rows are independent within and across domains. Observations are $x_e = \Sigma_e^{1/2} u_e$ with u_e standard Gaussian and $\Sigma_e = \mathrm{diag}(\lambda_j^e)$ diagonal with power-law spectra

$$\lambda_j^e \asymp j^{-\alpha_e}, \qquad \lambda_1^e = 1, \tag{1}$$

and throughout we abbreviate $a_j = \lambda_j^r$, $b_j = \lambda_j^h$.

Assumption 2 (Labels and mismatch)**.** $y_r = \langle \beta^*, x_r \rangle + \varepsilon_r$ in the robot domain; $y_h = \langle \beta^* + \delta, x_h \rangle + \varepsilon_h$ in the human domain. The *task mismatch* δ absorbs retargeting error, kinematic differences, and residual appearance-induced shifts; its *alignment* with the task is $c = \delta^\top \beta^*$. For each domain e and sample i, $\varepsilon_{e,i}$ has mean zero and variance σ_e^2, is independent of the design and of other samples, and has finite fourth moment. The fourth-moment assumption is used only for Cauchy–Schwarz and complementary-event bounds. The parameter β^* obeys an ellipsoid constraint $\sum_j j^{2s} \beta_j^{*2} \le 1$ with $s > 0$.

Assumption 3 (Mixture estimator)**.** Given n_r robot and n_h human samples and a mixing ratio $\lambda \in [0, 1]$, per-sample weights are $w_r = (1 - \lambda)/n_r$, $w_h = \lambda/n_h$. The main estimator is the *standard full-ridge* solution

$$\hat{\beta} = \left(w_r X_r^\top X_r + w_h X_h^\top X_h + \mu I\right)^{-1} \times \left(w_r X_r^\top y_r + w_h X_h^\top y_h\right), \qquad \mu > 0. \tag{2}$$

The diagonalized estimator, which retains only the diagonal of the sample Gram matrix, is analyzed separately as an interpretable special case. The coefficient μ is the ridge penalty and stands in for explicit or implicit regularization.

The evaluation criterion is the *robot-domain excess risk*, weighted by the robot geometry:

$$\mathcal{R}_T(\lambda) = \mathbb{E}\left\|\hat{\beta}(\lambda) - \beta^*\right\|_{\Sigma_r}^2. \tag{3}$$

Only errors in directions the robot actually visits are penalized — this weighting turns out to be essential for the phase structure in Sec. 4.

Full-ridge error and sample coupling. Let $G_e = X_e^\top X_e / n_e$ and $\widehat{A}_\lambda = (1 - \lambda) G_r + \lambda G_h + \mu I$. The exact error is

$$\hat{\beta} - \beta^* = \widehat{A}_\lambda^{-1}\big(- \mu \beta^* + \lambda G_h \delta + \eta_\lambda \big), \tag{4}$$

where $\eta_\lambda = w_r X_r^\top \varepsilon_r + w_h X_h^\top \varepsilon_h$. Thus shrinkage and task mismatch determine the population bias, while sample-Gram fluctuations and label noise contribute finite-sample corrections. Full ridge jointly uses the off-diagonal sample-Gram entries. The additional per-coordinate signal variance, or *label feedthrough*, of the diagonalized estimator is derived separately in the supplementary; it is not an extra independent noise source in Eq. (4). For example, with $\delta = 0$ and no label noise, the exact error is $-\mu \widehat{A}_\lambda^{-1} \beta^*$.

What the model keeps and discards. The model retains the three gap components that mixture practice manipulates — spectra (coverage), mismatch δ (embodiment/retargeting), and noise — and discards the rest (nonlinear observation maps, temporal structure, closed-loop compounding). Sec. 5 reports exact algebraic checks, population root and interval certificates, and within-model conditional Monte Carlo. Exploratory trained-model and assumption-stress protocols are recorded separately in the supplementary; policy-level validation is not claimed here. The diagonalized estimator is a separate simplified comparator to Assumption 3; its coordinatewise effective-noise analysis does not replace the full-ridge expansion. The full-ridge risk and its first two derivatives are expanded at fixed dimension in Theorem 3. The earlier diagonal-motivated full-ridge surrogate is retained only as a numerical comparator, not as the proved second-order object.

4. Global geometry and finite-sample selection

We first establish that the population mixing risk can have multiple valleys, then certify its global candidates, and finally study how finite samples shift or select the optima. Throughout, w denotes the mixture weight λ of Sec. 3, and $R_n(w)$ is the actual full-ridge target risk. Proofs and exact certificates are in the supplementary material.

4.1. Two coordinates suffice for multiple valleys

Taking the population limit of the full-ridge error in Eq. (4) gives the following risk; the derivation is provided in supplementary Theorems T1-b and T1-c. With $D_j(w) = (1-w)a_j + wb_j + \mu$,

$$R_0(w) = \sum_j a_j \frac{(wb_j\delta_j - \mu\beta_j^*)^2}{D_j(w)^2}. \tag{5}$$

Proposition 1 (Minimum dimension for multiple valleys)**.** *For the population risk in Eq. (5), when $p = 1$, $R_0'(w) = L(w)/D(w)^3$ with affine L, so there is at most one non-degenerate local valley. In contrast, a certified two-dimensional example satisfying Assumptions 1–3 has three simple interior stationary points*

$$w_1 \approx 0.264842, \quad w_2 \approx 0.727011, \quad w_3 \approx 0.869112, \tag{6}$$

which are respectively a local minimum, a local maximum, and a local minimum. Interval risk comparison proves that w_1 is the unique global minimizer. Thus $p = 2$ is the minimum dimension for multiple valleys.

Figure 1 (left) shows this certified double-valley risk curve. Varying its second-coordinate weight t gives valley-pair creation at $t_1 \approx 0.9375848$ and annihilation at $t_2 \approx 1.2003419$. Figure 1 (right) tracks the risks at the stationary points as t varies. Between these critical values, the global optimum switches branches, and at some $t_* \in [1.06085925, 1.06085926]$ the two local minima have equal risk: two different weights are global minimizers.

4.2. Certifying global population optima

Multiple valleys require a comparison of all candidate optima. Let $L_j(w)$ be the affine numerator of the derivative of $(wb_j\delta_j - \mu\beta_j^*)^2/D_j(w)^2$. Then $R_0'(w) = \sum_j a_j L_j(w)/D_j(w)^3$. Clearing the positive denominators gives

$$P(w) = \sum_j a_j L_j(w) \prod_{k\neq j} D_k(w)^3 = 0, \qquad \deg P \le 3p-2. \tag{7}$$

For fixed parameters, isolating all real roots of P in $(0,1)$ and comparing them with $w = 0, 1$ certifies the global minimizers without assuming convexity or unimodality. Degenerate roots and flat cases are handled separately.

Zero mismatch gives a tractable special case in which the shape itself determines how to find the optimum.

Proposition 2 (Log-convexity and flatness at zero mismatch)**.** *Suppose $\delta = 0$ and let $\mathcal{J} = \{j : a_j\beta_j^{*2} > 0\}$. Then R_0 is log-convex on $[0,1]$. If some $j \in \mathcal{J}$ has $a_j \neq b_j$, then R_0 is strictly log-convex and has a unique minimizer. If every $j \in \mathcal{J}$ has $a_j = b_j$, or $\mathcal{J}$ is empty, then R_0 is constant and every weight is optimal. In the strictly log-convex case, the signs of $R_0'(0^+)$ and $R_0'(1^-)$ determine whether the unique optimum is at zero, one, or an interior point.*

Thus the same model includes both a certified multivalley example and a zero-mismatch regime with a unique optimum or a flat risk. These population cases provide distinct starting points for finite-sample analysis.

4.3. From population geometry to finite samples

To relate the certified population candidates to actual-risk optima, we need uniform control of the risk, its slope, and its curvature.

Theorem 3 (Second-order full-ridge risk, T1-c)**.** *Consider the standard full-ridge estimator under the Gaussian design and noise model, with fixed p, fixed $\mu > 0$, and fixed $\rho = n_h/n_r$. Then, uniformly for $w \in [0,1]$,*

$$R_n(w) = R_0(w) + \frac{R_1(w)}{n_r} + o(n_r^{-1}), \tag{8}$$

where R_0 and R_1 are explicit matrix coefficients derived in the supplementary. The first two derivatives satisfy the same uniform expansion.

The expansion applies to actual full ridge at fixed dimension. Its uniform derivatives preserve positive local curvature for sufficiently large samples and permit the following three conditional selection results.

Theorem 4 (Nondegenerate interior optimum)**.** *Suppose R_0 has a unique global minimizer $w_0 \in (0,1)$ with $R_0''(w_0) > 0$. Then, for all sufficiently large n_r, R_n has a unique global minimizer w_n^* on $[0,1]$, with*

$$w_n^* = w_0 - \frac{R_1'(w_0)}{n_r R_0''(w_0)} + o(n_r^{-1}),$$
$$R_n(w_n^*) = R_0(w_0) + R_1(w_0)/n_r + o(n_r^{-1}). \tag{9}$$

Consequently, $\mathcal{G}_n = R_n(0) - R_n(w_n^*)$ satisfies $\mathcal{G}_n = \mathcal{G}_0 + \mathcal{G}_1/n_r + o(n_r^{-1})$, where $\mathcal{G}_0 = R_0(0) - R_0(w_0)$ and $\mathcal{G}_1 = R_1(0) - R_1(w_0)$.

Theorem 5 (R_1 selects among multiple population optima)**.** *Suppose R_0 has exactly two global minimizers $0 < w_1 < w_2 < 1$, both with positive local curvature. If $R_1(w_1) < R_1(w_2)$, then for all sufficiently large n_r the global optimum is attained only near w_1, and $w_n^* \to w_1$. The risk gap*

between the two local branches is $(R_1(w_2)-R_1(w_1))/n_r+o(n_r^{-1})$. *If the first-order values tie, this expansion does not select a branch.*

Theorem 6 (Flat R_0 at matched spectra and zero mismatch)**.** *Let* $\Sigma_r = \Sigma_h = \Sigma$ *and* $\delta = 0$. *Then* R_0 *is constant and there are explicit nonnegative coefficients* A, B *such that*

$$R_1(w) = A(1-w)^2 + Bw^2. \tag{10}$$

If $A+B > 0$, *then* R_n *is strictly convex for sufficiently large* n_r, $w_n^* \to w_1^* = A/(A+B)$, *and*

$$R_n(w_n^*) = C + \frac{AB}{A+B}\frac{1}{n_r} + o(n_r^{-1}),$$
$$\mathcal{G}_n = \frac{A^2}{A+B}\frac{1}{n_r} + o(n_r^{-1}), \tag{11}$$

where $C = R_0$ *and* $\mathcal{G}_n = R_n(0) - R_n(w_n^*)$. *If* $A = B = 0$, *then* $\beta^* = 0$ *and both label-noise variances vanish, so* $R_n(w) = 0$ *for every sample size and weight: all weights are optimal and* $\mathcal{G}_n = 0$ *exactly.*

Implications for mixing. A positive zero-mixing derivative cannot by itself justify discarding the source, and a unique global minimizer does not imply unimodality. Global population certification identifies the candidates; the uniform expansion then determines local shifts and first-order selection under the conditions above. It does not classify the entire finite-sample landscape at an arbitrary sample size.

Further consequences. The supplementary material derives matched-spectrum gain formulas, a first-order exchange approximation, and a dosage rule for an explicitly defined quadratic surrogate. The dosage rule concerns that surrogate; the exchange approximation is restricted to first-order risks and a fixed-regularization baseline.

5. Numerical validation

Validation follows the theoretical sequence: population valleys and global certification, then finite-sample risk and weight corrections. Exact algebra, root isolation, and interval comparisons certify population objects; conditional Monte Carlo checks actual finite-sample full-ridge risk. The fixed-candidate comparison below is reported separately from validation of optimized finite-sample branches.

Global-shape certificates. For the two-dimensional multimodal example, exact real-root isolation gives three simple interior stationary points at 0.2648422643, 0.7270105700, and 0.8691124964. Rational interval evaluation proves that the first is the unique global minimum, with risk interval $[1.9948444989, 1.9948445856] \times 10^{-5}$; the second local minimum has a strictly higher interval. The example satisfies exact power-law spectra and the task ellipsoid constraint. Independent NumPy roots differ from the exact roots by at most 1.04×10^{-14}.

For the one-parameter family, the discriminant has two positive roots $t_1 \approx 0.9375848094$ and $t_2 \approx 1.2003419385$. Sturm counts confirm one, three, and one interior roots in the three regions. At $t = 1$ the first branch is globally optimal, while at $t = 1.1$ the third branch wins. Inside $[1.06085925, 1.06085926]$, the branch-risk difference changes sign while all other candidates remain strictly above, proving the existence of two global minimizers. The numerical estimate is $t_* \approx 1.0608592591$.

Full-ridge expansion and optimum checks. The fixed-dimension second-order risk expansion is checked in 22 design-level cells over $p = 1, 2, 20$; all simulated means are within 2.13 Monte Carlo standard errors. Four optimized-risk checks are within 1.59 standard errors. For the two-dimensional aligned example, $w_0 = 0.3708597$, $R_0''(w_0) = 0.13996$, and the predicted scaled weight shift is -2.29329; 200,000 conditional-design replicates at $n_r = 4000$ give -2.29688 ± 0.07803. Actual-E2 paired checks at $n_r = 1000$ give derivatives -0.000435 ± 0.000137 and 0.000469 ± 0.000137 on the two sides of the first-order prediction.

Selection from a flat population risk. For matched spectra and zero mismatch, 200 random configurations give maximum R_1 quadratic-identity error 1.82×10^{-12} and maximum flat-R_0 error 3.55×10^{-15}. Four conditional-risk gain checks at $n_r = 1000, 4000$ are within approximately 1.18 standard errors.

Risk ordering at two fixed population candidates. For the multiple-population-optimum example, we evaluate the finite-sample risk at the two *fixed population weights*. Here $R_1(w_1) = 1.56977$ and $R_1(w_2) = 5.14388$; at $n_r = 1000$, the first-order fixed-point gap is 0.00357412 and 20,000 conditional-risk replicates give $0.00357638 \pm 0.00000078$ (mean $\pm$ one standard error). This supports the ordering of these two fixed candidates; it does not locate finite-sample local optima or validate branch persistence. The first-order weight displacement at w_1 is about 13.91, outside $[0, 1]$, so this sample size is not a small-displacement check of Theorem 5.

6. Discussion

What global geometry changes. Two coordinates already permit competing local minima, so a locally optimal

mixture need not minimize target risk globally. For specified model parameters, root isolation and risk comparison identify the global population candidates. Uniform risk and derivative control then connect these candidates to finite-sample shifts and asymptotic selection under the stated non-degeneracy conditions. This separates the question of where all population optima lie from the question of how sampling changes their preference.

Limitations. The analysis uses fixed dimension, Gaussian covariates, a shared spectral basis, and fixed regularization. The log-convex classification is restricted to zero mismatch; the branch diagram covers a specific two-coordinate family. A general finite-sample shape classification is not established, and the fixed-candidate experiment does not verify persistence of two finite-sample local minima. Global certification assumes specified population parameters; robustness to estimating them and validation on native visual or robot-policy tasks remain open.

Outlook. The next steps are to quantify when finite samples preserve multiple valleys, evaluate the actual optimized branches in that regime, and study sensitivity to estimated parameters and tuned regularization. Native cross-embodiment experiments should then test whether the predicted weight choices improve both open-loop risk and closed-loop performance.

Conclusion. We established that two coordinates suffice for multivalley mixing risk, certified global population optima algebraically, and connected these structures to finite-sample optimal selection through uniform risk expansions. The resulting distinction between local and global choice is the central implication for auxiliary-data mixing under the stated model.

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